

BAAZ JKBOSE



2026 CLASS-11TH

Chemical Control and Coordination

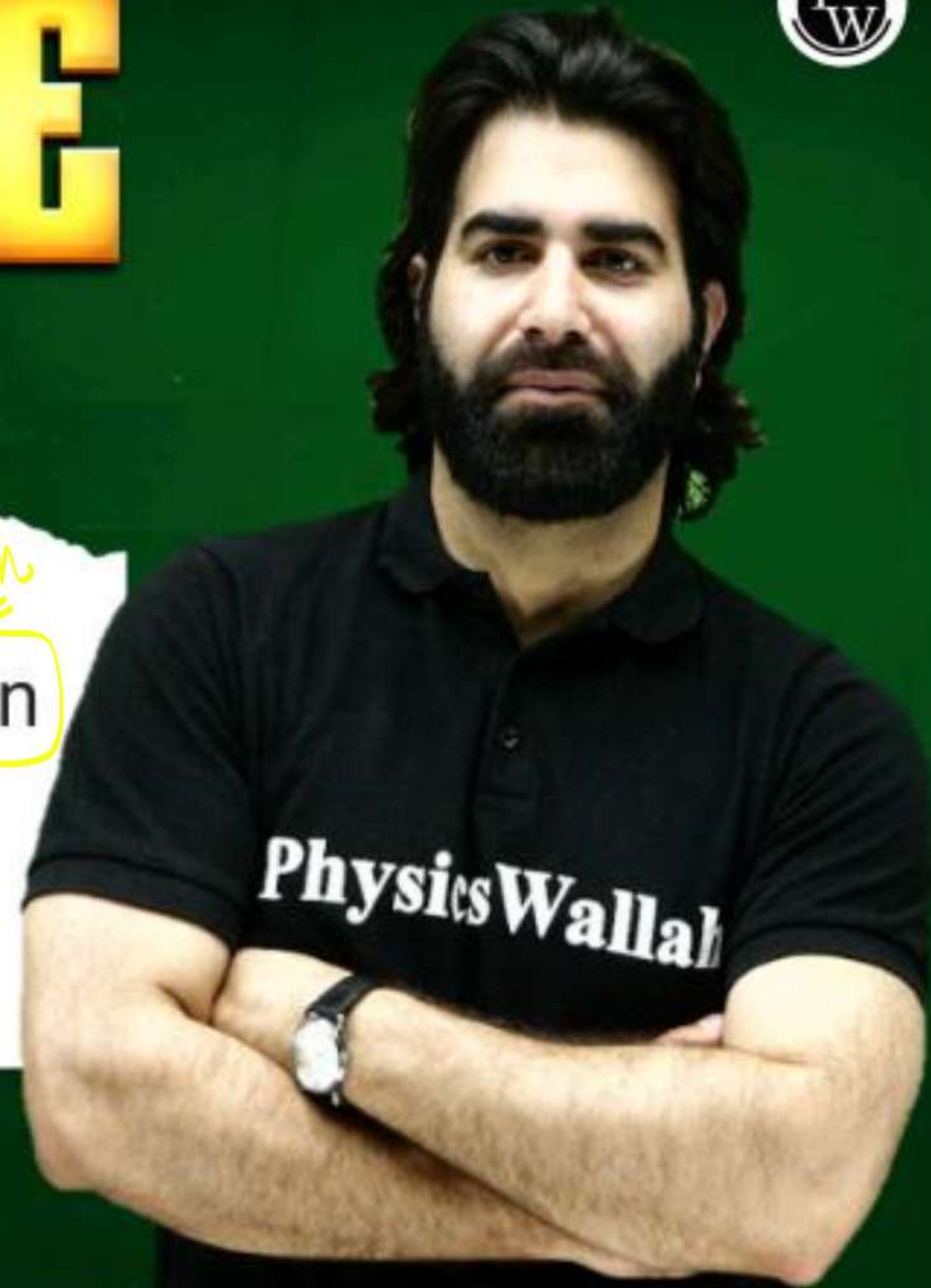
Common

or
Chemical Coordination and Integration

One Shot

Zoology (biology)

Dr Adil Altaf sir



Topics To Be Covered

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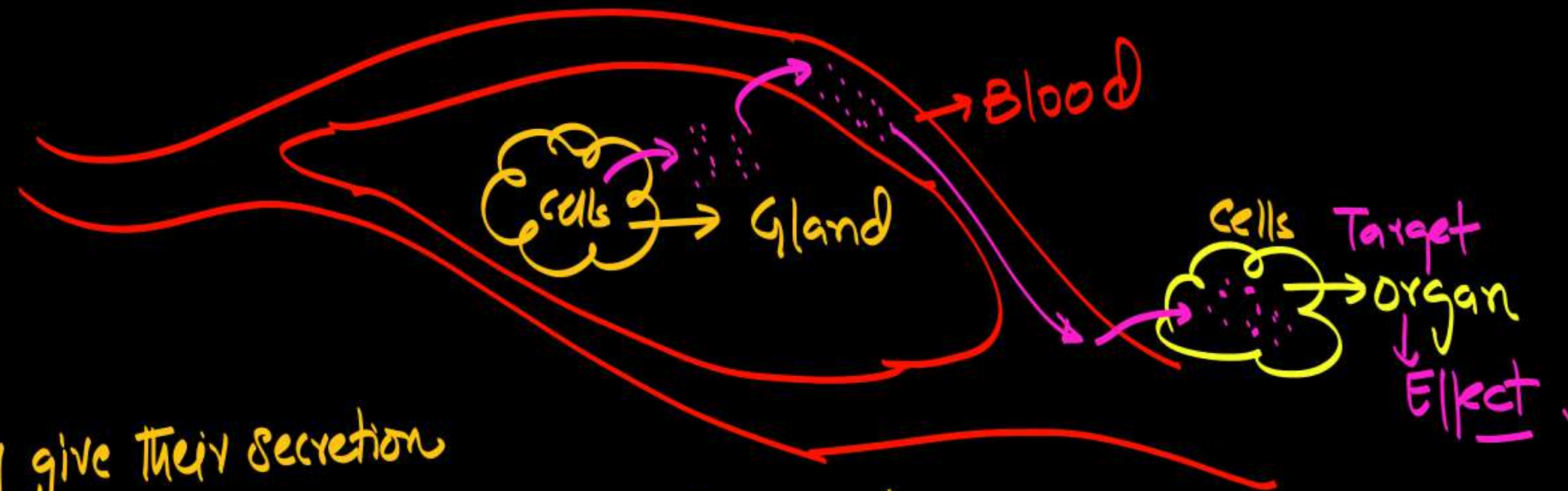
Saraa Chapter

- Introduction
- Hormones
- Endocrine Glands.

→ Chemical Control and Coordination

- Endocrine Glands → Any cell / organ / Tissue — Secretory in Function

↳ These are those glands which Do not have Ducts.



↳ They give their secretion

Directly into Blood through Surrounding Fluid.

↳ Their secretion contains Hormones



Hormone

↳ These are intercellular chemical messengers, which are non-nutritious and produced in between cells.
 (Between)

Trace Amounts → Produced in Micro & Nanograms

↳ Nature of Hormone (Chemically)

→ protein / polypeptide - Pituitary Hormone, Hypothalamus, Calcitonin, PTH, Insulin, Glucagon etc



→ Steroid - Adrenal Cortex Hormone, Sex Hormones

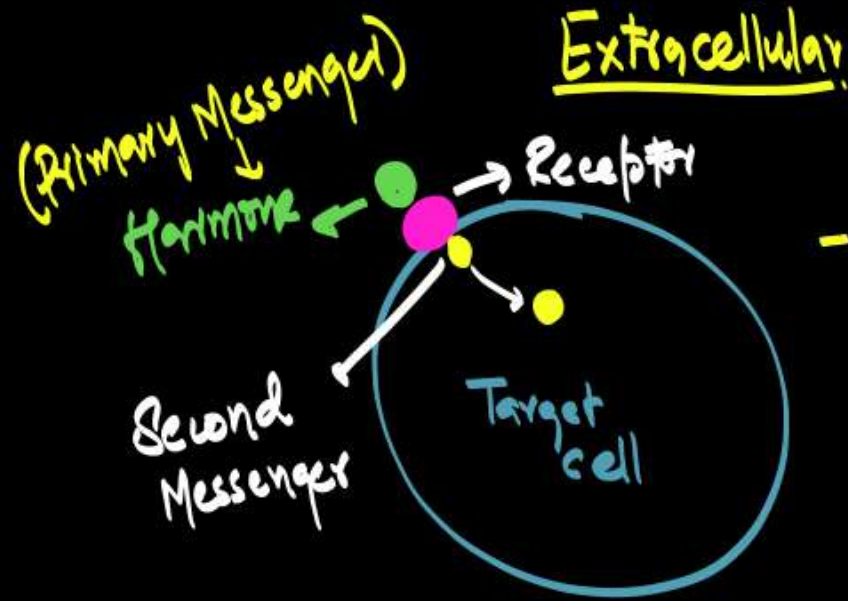


→ Derivative of Amino Acid (Tyrosine) → Adrenal Medullary Hormone (Adrenaline, Nor Adrenaline)
 → Thyroid Hormone (T₃, T₄)
 (Iodothyronine)

→ Mechanism of Hormone Action.

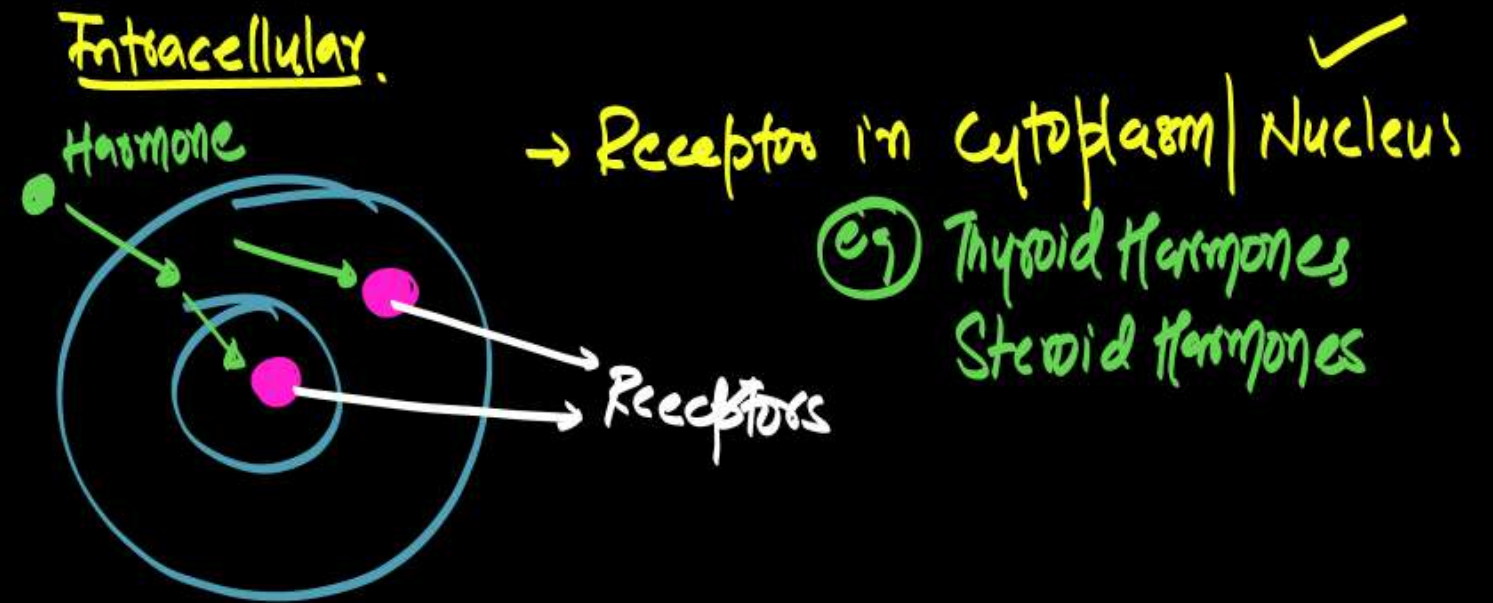
Hormone attaches to Receptors on target cells and forms Hormone Receptor Complex (HRC)

H.R.C



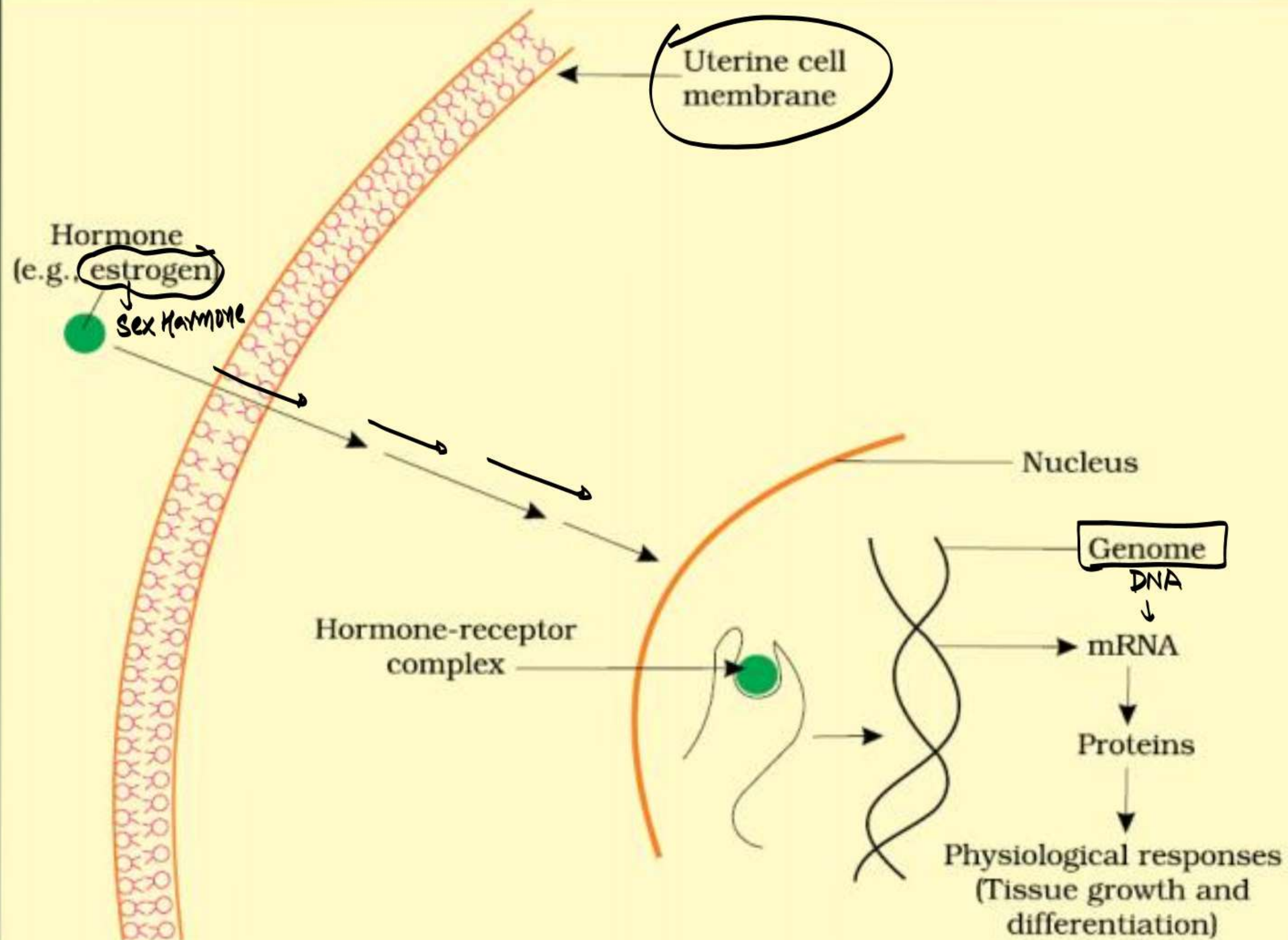
- Receptor on plasma Membrane.
e.g - Protein/polypeptide
- Adrenal Medullary.
→ Here Second Messenger is Required.

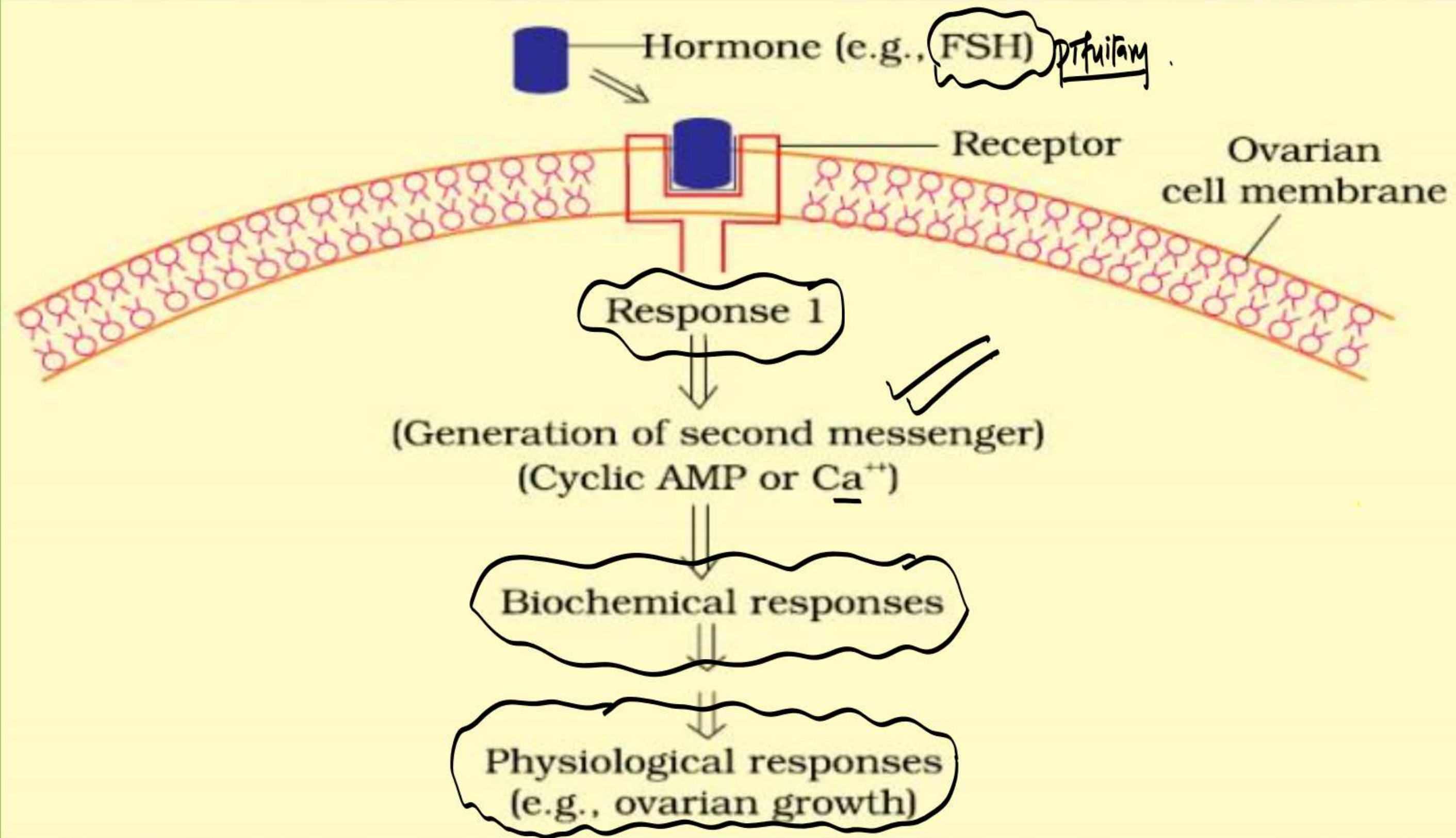
e.g cAMP, IP₃, Ca⁺⁺ etc



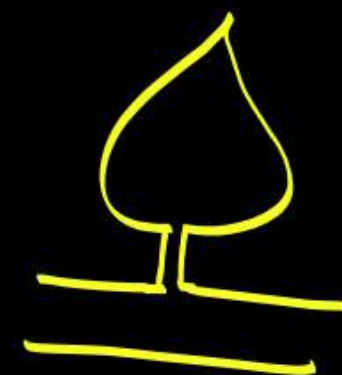
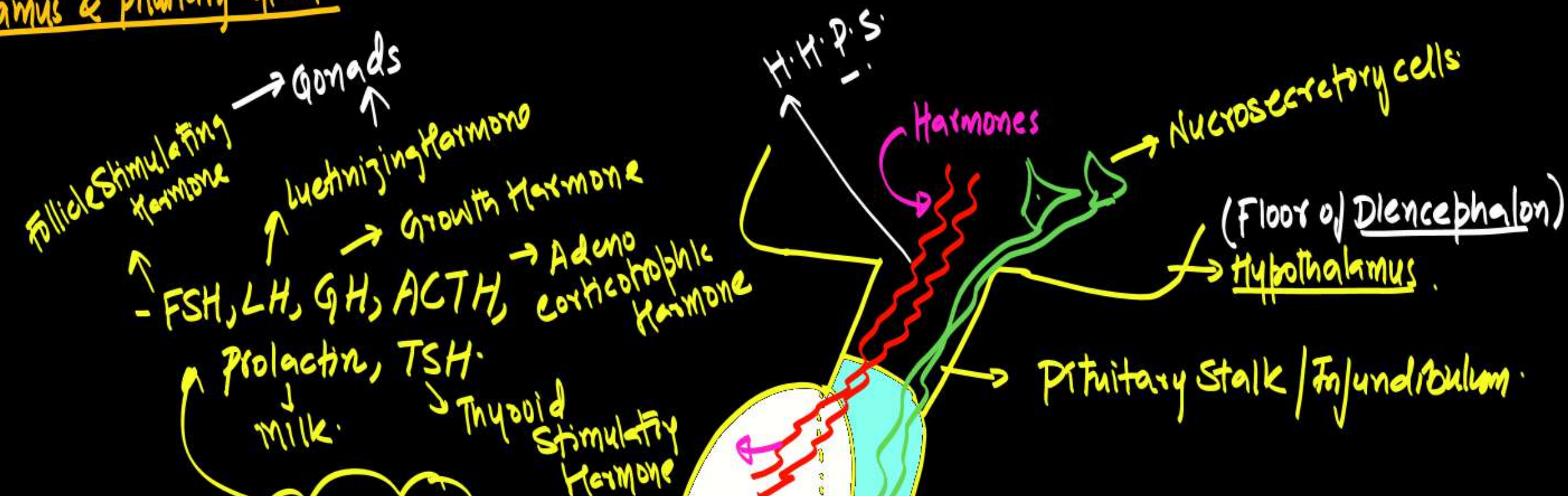
e.g Thyroid Hormones
Steroid Hormones

Hormone after Binding to Receptors Finally Brings Biochemical changes in cell leading to Physiological change.





Hypothalamus & Pituitary Gland.



Anterior pituitary

Posterior pituitary

Controls all other endocrine glands of Body.

Pituitary Gland (Master Endocrine Gland)

↳ Also known as Hypophysis

↳ Two parts - Anterior pituitary (Adenohypophysis) - posterior pituitary (Neurohypophysis)

↳ located in bony cavity - Sella turcica of Sphenoid Bone.



- (e.g.) to Release GH from pituitary, Hypothalamus sends Growth Hormone Releasing Hormone
to Inhibit " " " , " " " " " " " " Inhibitory "
GH IH / Somatostatin

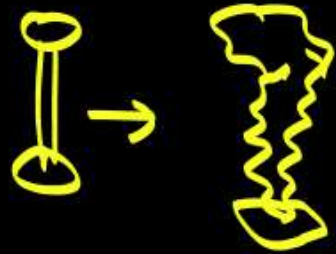
- Hypothalamus controls post. pituitary by Neurons.
Actually post. pit. hormones are produced by hypothalamus and stored in post pituitary.

Some Disorders of Pituitary Gland:

① Pituitary Dwarfism - low GH During Growth Years.

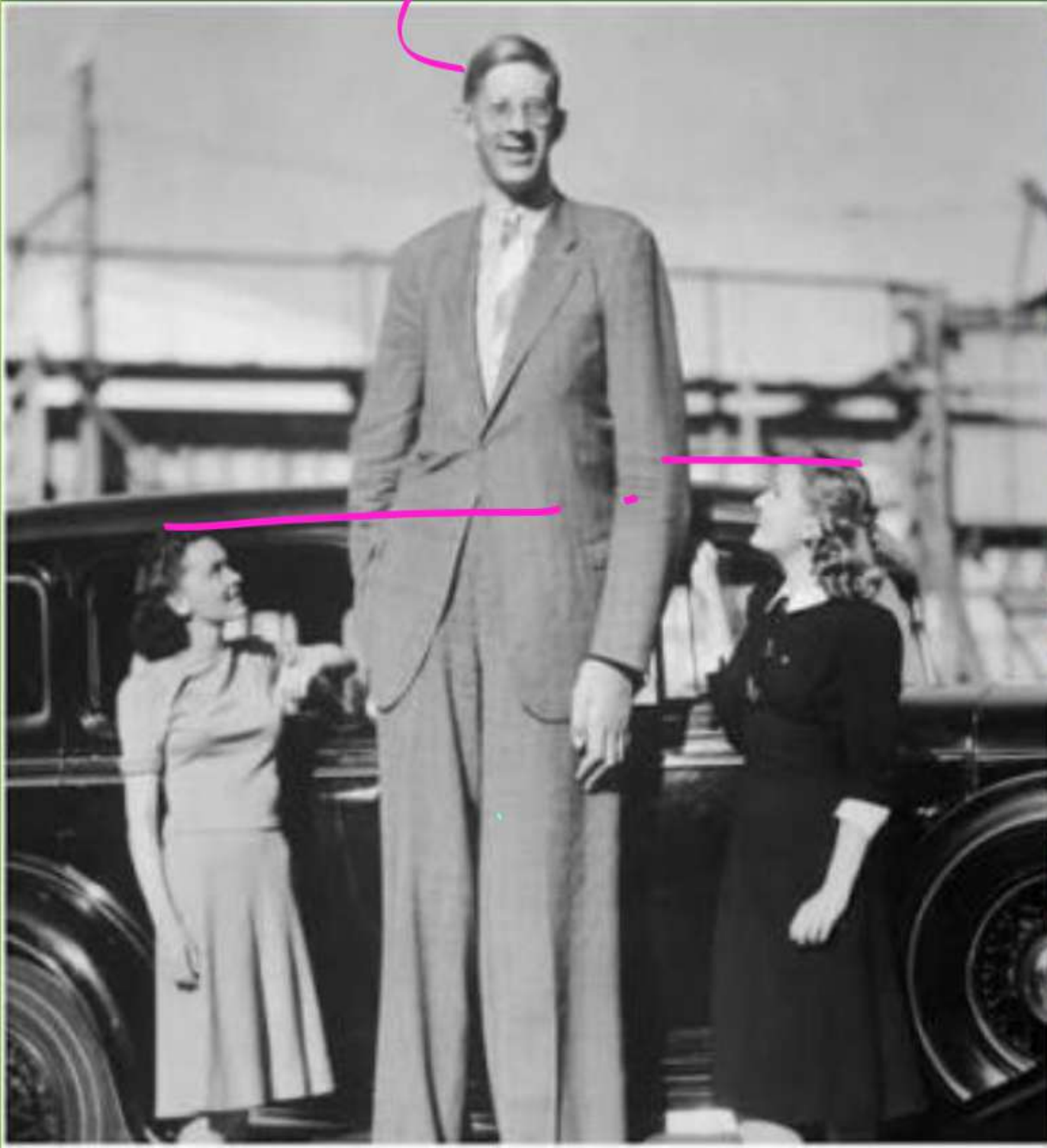
② Gigantism - More GH During Growth Years.

③ Acromegaly - More GH in Middle Age, Abnormally Thickening Bones, Face, Hand etc



④ Acromegalic gigantism $\rightarrow$ H/W.

Gigantism



Eyeball
Bulged
out

Dwarf



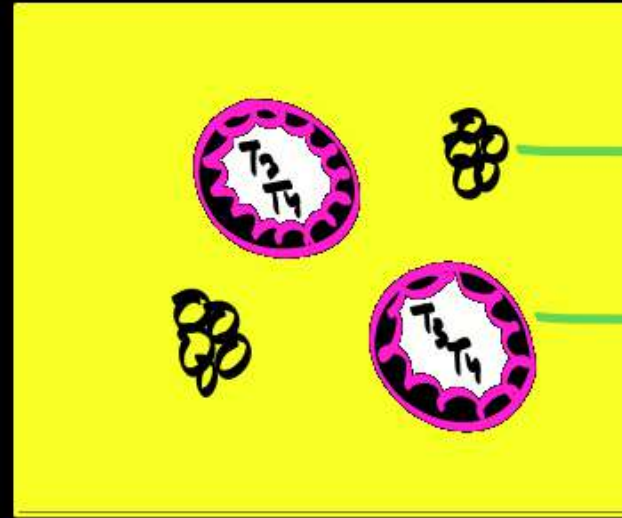
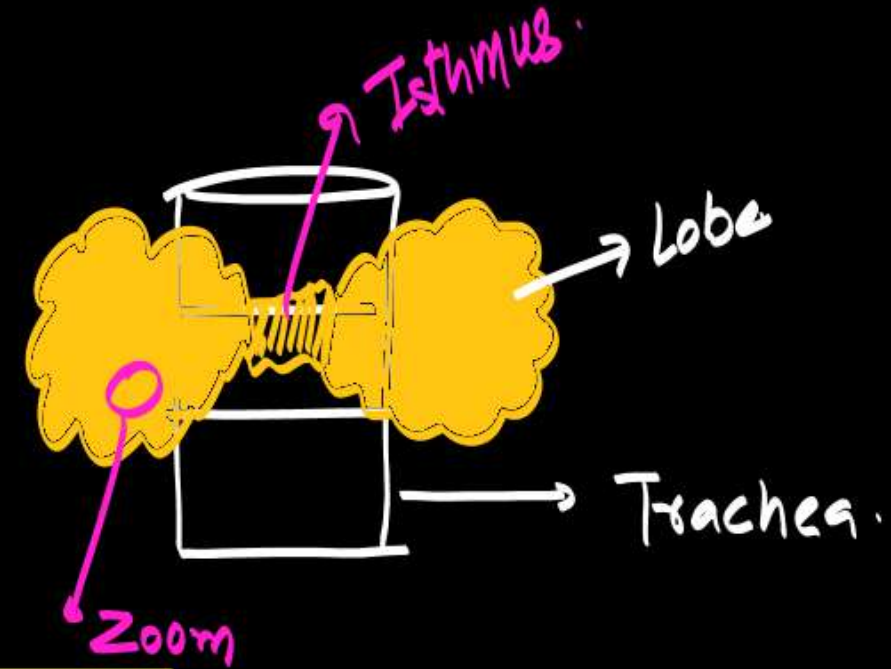
Goutte

Before vs After Excess Growth Hormone



Thyroid Gland.

- ↳ It is located on the trachea
- ↳ It is Bilobed / Butterfly shaped Gland.
- ↳ Inside of this Gland.



Para Follicular / C-cells → Calcitonin.

Follicle → produce T₃, T₄.

T₃ vs T₄

→ Triiodothyronine

→ Less Quantity

→ More Effect (shorter Duration)

Tetraiodothyronine (Thyroxine)

- More Quantity

- less effect (longer Duration)

Function of T_3 / T_4

- ① Maintain / Increase (BMR) - BMR = Basal Metabolic Rate
 It is Minimum Rate of Metabolism required to keep a person Active at Rest. (Average 1600 kcal / Day)
- ② Involved in Metabolism of protein, Carbohydrates, lipids
- ③ Helps in RBC Formation.
- ④ Imp for Normal Menstrual Cycle.
- ⑤ Maintain BP & HR.

Disorders

- Hypert thyroidism
- More T_3, T_4
- TSH low



Hypothyroidism → 1

- Less T_3, T_4
- TSH high

e.g. ① Graves Diseases

[Exophthalmic Goitre]

↳ Enlargement
of Thyroid Gland.

- Autoimmune Disease

② - Also Be due to thyroid tumours

→ In Pregnancy. It can lead
Stunted growth (cretinism).

- Low IQ.
- Deafness etc.

→ Iodine Deficiency

Parathyroid Gland.

- four Round glands located on posterior side of thyroid lobes
- These glands produce parathormone (PTH)
- PTH is opposite to Thyroid Calcitonin (TCT)



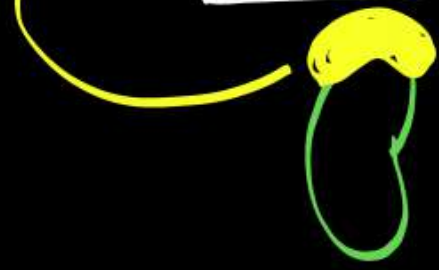
PTH.

- Increases Blood Ca^{+2} level (Hypercalcemic).
- ① ↑ Absorption from GIT.
- ② ↑ Reabsorption from kidney tubules.
- ③ ↑ Bone Resorption.

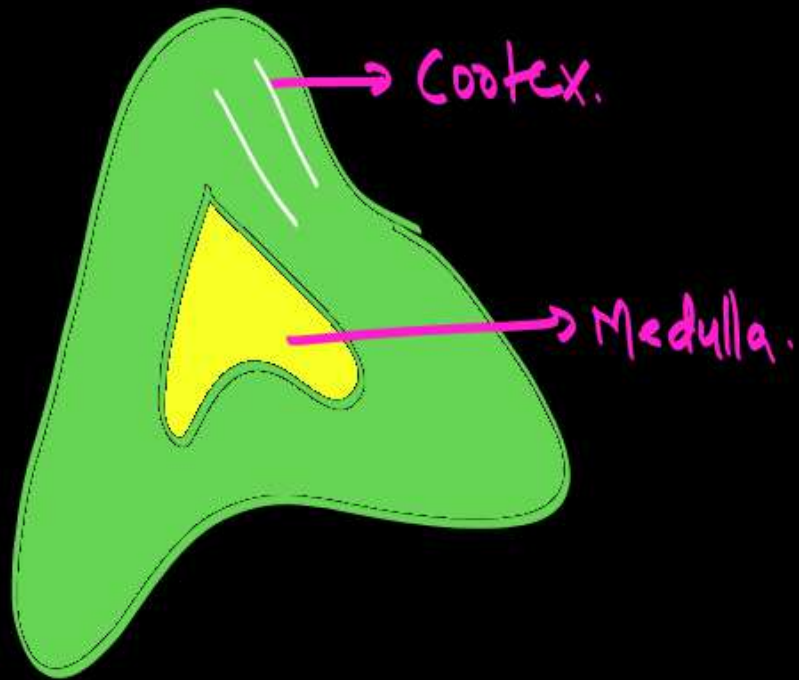
TCT.

- Decreases Blood Ca^{+2} level (Hypocalcemic).
- ↓ Absorption from GIT.
- ↓ Reabsorption from tubules.
- ↑ Bone Deposition.

- Adrenal Glands (Emergency Glands)
- located on the upper pole of kidney.



→ Inside these Glands have



GFR:

Cortex

→ Cortex produces Steroid Hormones.

- It has three Sub-Regions

① Zona Glomerulosa → Produce Mineralocorticoid
e.g. Aldosterone (Main)

- Maintain ionic Balance
- cause Na⁺ Reabsorption & K⁺ Excretion.

② Zona Fasciculata - Produces - Glucocorticoids
e.g. Cortisol (Main)

- Elevate Blood Glucose level.
- Metabolism of Carbs, Protein (Proteolysis), Lipid (Lipolysis)
- Anti-inflammatory, RBC Formation (Helps).

③ Zona Reticularis

- produces Sex Corticoids
- e.g. Androgens in Females normally cause growth of Pubic & Axillary Hair.

→ Adrenal Medulla produces - Adrenaline (Epinephrine), Nor Adrenaline (Norepinephrine), Dopamine

→ It produces it during Emergency conditions (Flight / Fight Situation).

→ These hormone do Sympathetic Stimulation. { Chased by Lion }

like - increase HR.

- Dilation of pupil

- ↑ Glucose level

- ↑ Blood pressure

- Dilation of Bronchi Bronchioles

- Erection of Hair

- Relaxation of Bladder

- ↓ Digestion.

Pancreas

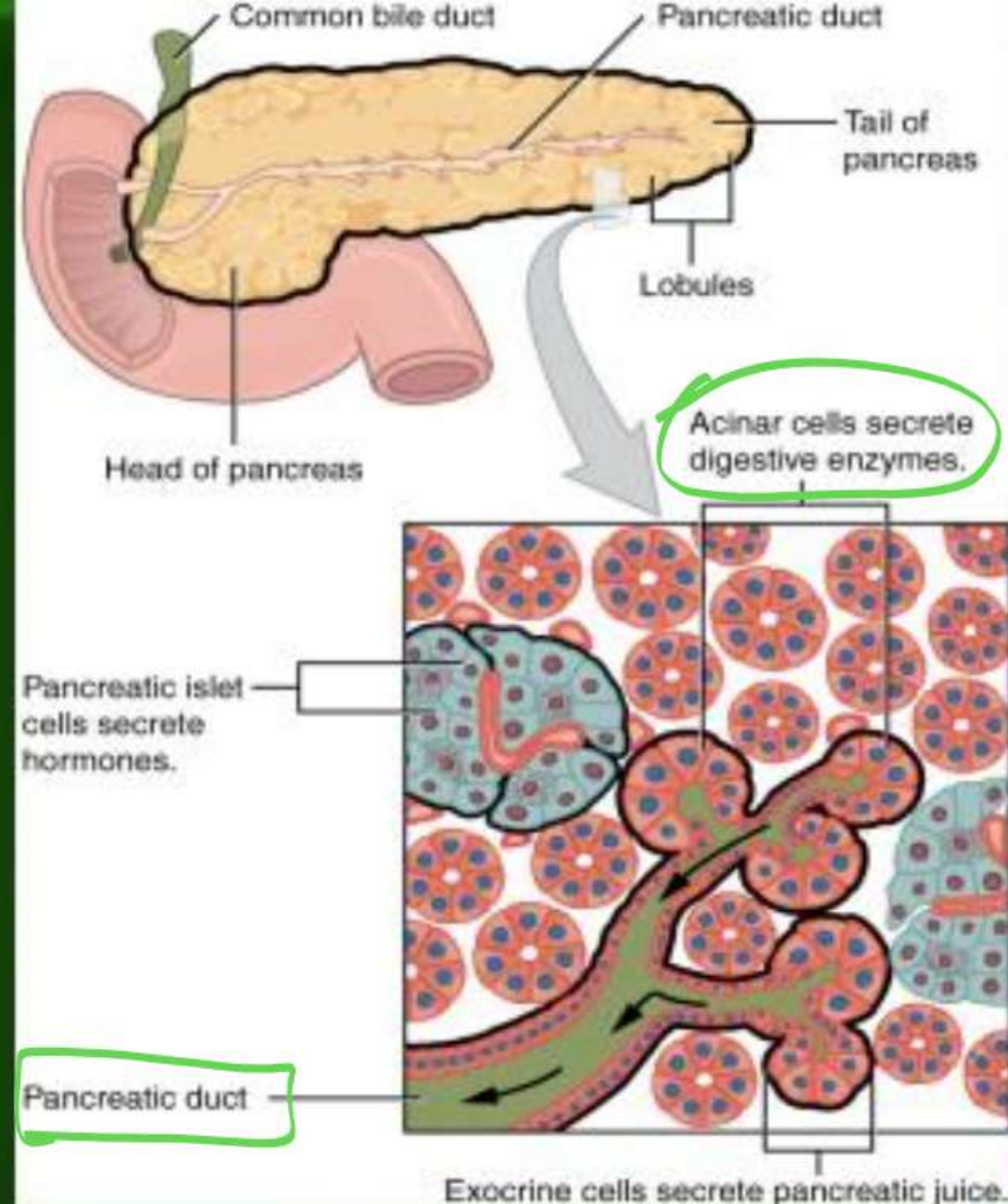
Heterocrine Gland

- Exocrine part = Pancreatic Acini
- Endocrine part = Islets of Langerhans



Four types of cells in Islets

- α -cell (Alpha) → Glucagon
 - β -cell (Beta) → Insulin
 - δ -cell (Delta) → Somatostatin
 - PP-cell → Pancreatic polypeptide
- } opposite (Antagonistic)
} I/W



Glucagon

- Released when Blood glucose level Falls.
- Increases Blood glucose level
(Hyperglycemic Hormone)

→ By:

① Glycogenolysis

Glycogen → Glucose.

② Gluconeogenesis

Formation of Glucose from Non
Carbohydrate Sources e.g. Amino
Acid.
- Fatty Acid

Insulin

- Released when Blood glucose level Rises
- Decreases Blood glucose level.
(Hypoglycemic Hormone).

- By:

① Glycogenesis

Glucose → Glycogen.

② Cellular Intake of Glucose

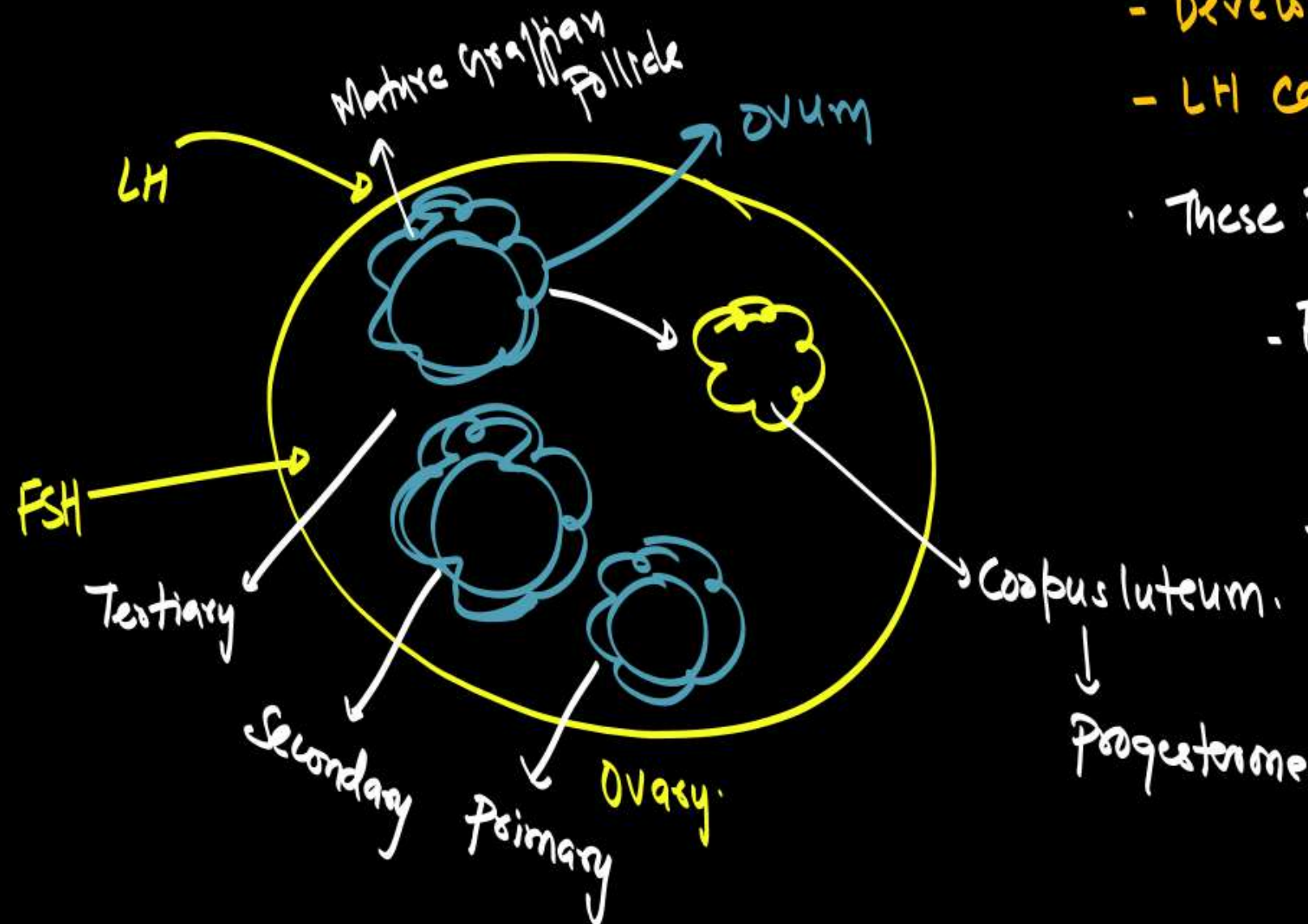
→ Oppose.

→ Deficiency of Insulin - Diabetes mellitus

Gonads - Gamete Forming organs

Includes Testis & ovaries → Overview

→ Ovaries → contain follicles having developing ovum.



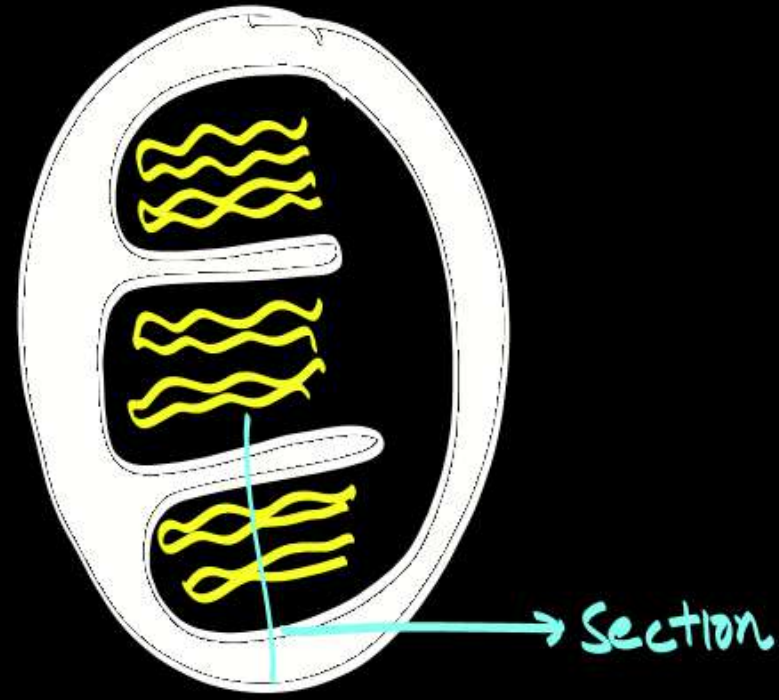
- FSH causes stimulation of follicle
- Develops them till mature Graafian follicle.
- LH cause Release of ovum.

These follicles Release

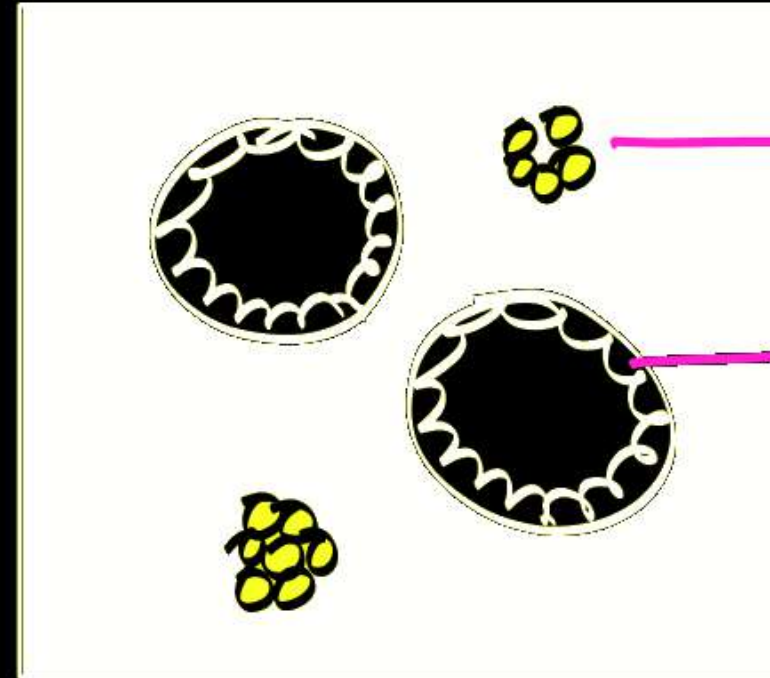
- Estrogen - Growth of Endometrium of Uterus
- Female Secondary sex characters
- Development of Reproductive organs.
- Progesterone.
 - Growth of Endometrium
 - Development of Alveoli of M. Glands.
 - ↓
 - milk.

Testes

- Testosterone - Spermatogenesis
- Development of Secondary Sexual characters
- Development of Reproductive organs.



Testis contain
Seminiferous tubule



Testosterone.



Leydig cells
(Interstitial cells)

LH



germinal epithelium

FSH



↓
Sperms.

Some other glands

① Pineal gland → produces

Melatonin

↓ Dark.

- For Diurnal Cycle

- 24 hr Rhythm cycle eg Sleep Wake cycle.

- Pigmentation

- Menstrual cycle.

② Thymus Gland

→ Located btw Sternum & Heart

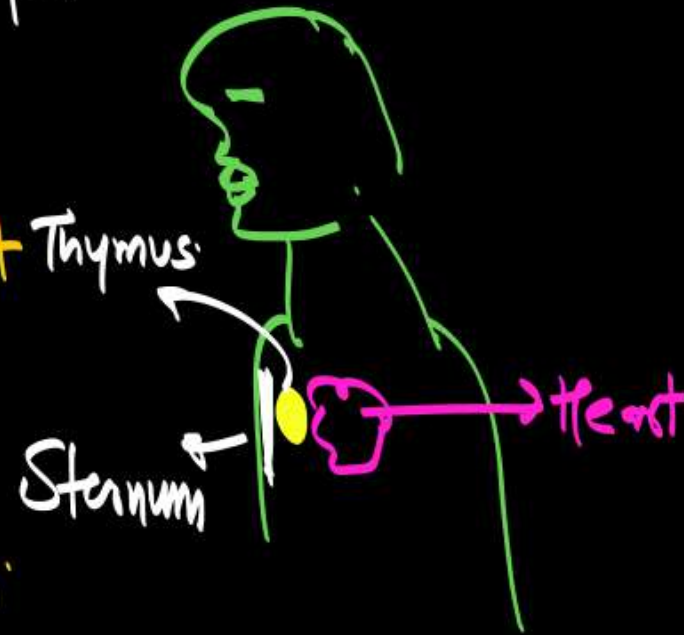
→ produces hormone thymosin

- Imp For Maturation of lymphocytes.

- Imp for Immunity

- Max Size Puberty

- After that it Diminishes.



④ Heart — ANF Atrial Natriuretic Factor

⑤ Kidney → Erythropoietin → RBC Formation

⑥ GIT → Gastrin, Secretin, GIP, CCK

Cholecystokinin
pancreozymin → Contraction of Gall Bladder
and pancreas.

- From Stomach

- ↑ Gastric Juice (HCl, pepsinogen)

- ↑ Gastric Mobility

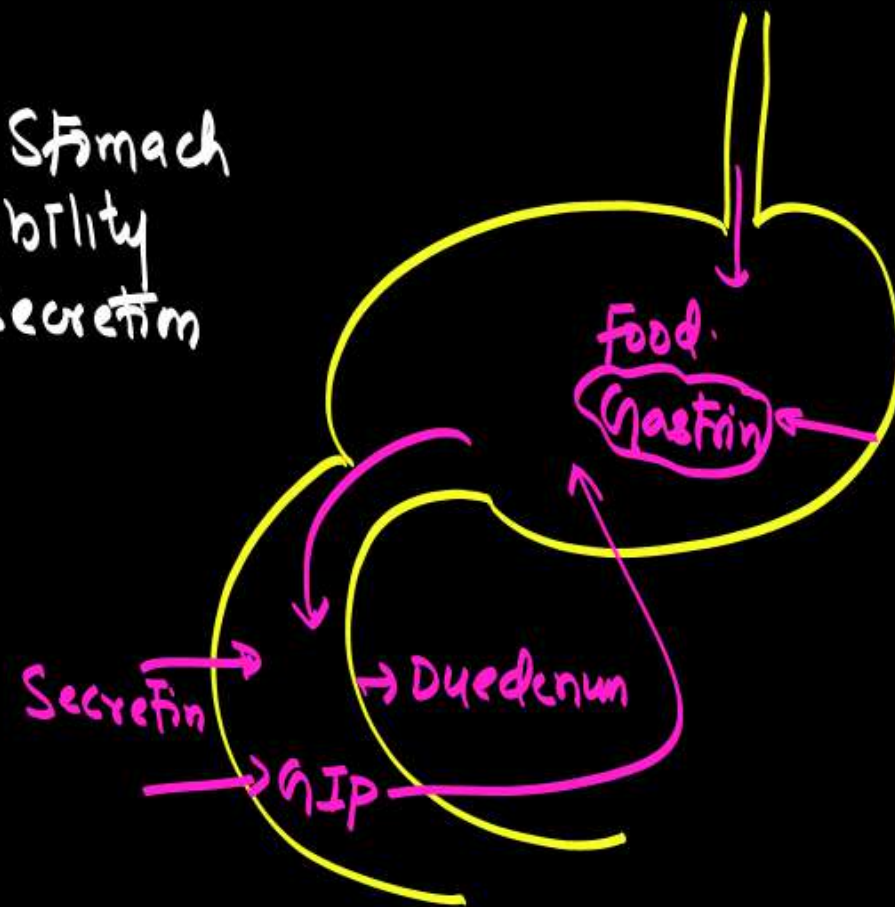
- From Intestine

- Acts on Pancreas - ↓ Secretion

- ↑ Bicarbonates

Acts on Stomach

- ↓ Mobility



Thank
You